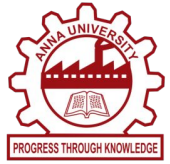




Incubating Minds,
Catalyzing Careers

AGI1242 – MACHINE LEARNING TECHNIQUES REVIEW-2



Department of ARTIFICIAL INTELLIGENCE & DATA SCIENCE Academic Year: 2026 – 2027 (Odd Semester)

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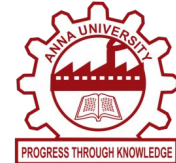
**INCUBATING MINDS
CATALYZING CARRERS**

**K.RAMAKRISHNAN
COLLEGE OF TECHNOLOGY**

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Automated Pneumonia Detection from Chest X-rays Using CNN

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PRESENTATION OVERVIEW



- **Project Title**
- **Objective**
- **Problem Identification**
- **Flow Diagram**
- **Literature Survey**
- **Research Methodology**
- **Compare algorithm**
- **Implementation idea**
- **Modules**
- **Conclusion**



OBJECTIVE



- To reduce the diagnosis time by enabling faster, AI-assisted screening of X-ray images.
- To evaluate the model's accuracy in correctly identifying pneumonia cases.
- To provide a simple and reliable tool that can support doctors in clinical decision-making.
- To make pneumonia detection accessible even in settings with limited radiology expertise.
- To build an automated system that classifies chest X-ray images as "Pneumonia" or "Normal" using a pre-trained deep learning model.

PROBLEM IDENTIFICATION

- Lack of an automated system makes it difficult to quickly screen chest X-rays for pneumonia in high-patient-volume hospitals.
- Human error and fatigue can lead to missed or delayed pneumonia detection, especially in emergency cases.
- Manual diagnosis by radiologists is time-consuming and depends heavily on individual expertise and experience.
- Many healthcare centers, especially in rural areas, lack access to expert radiologists for quick diagnosis.
- Absence of a standardized comparison of AI models makes it hard to choose the right one for real-world deployment.
- Traditional diagnosis methods do not scale well when handling a large number of X-ray images at once.

FLOW DIAGRAM

Data Preparation and Model Building



Training, Evaluation and Prediction





LITERATURE SURVEY



[Literature survey google sheet link](#)

METHODOLOGY

1. Dataset Preparation

Curate chest X-ray images from reliable, publicly available medical datasets. The images are carefully preprocessed through resizing and normalization, with data augmentation applied to enrich variability and strengthen model generalization.

2. Model Selection

A custom Convolutional Neural Network is built from scratch (no pre-trained weights). Stacked Conv2D + MaxPooling blocks with increasing numbers of filters extract features from the X-ray, followed by a Flatten layer, fully connected Dense layers and a final output layer that classifies the image as Normal or Pneumonia.

METHODOLOGY

3. Training and Fine-tuning

Employ transfer learning to refine the models on labeled chest X-ray data. Training is guided by an appropriate loss function, such as categorical cross-entropy, and optimized using algorithms like Adam or SGD to ensure stable convergence and improved performance.

4. Evaluation and Comparison

Assess each model using comprehensive performance metrics, including Accuracy, Precision, Recall, F1-Score, and ROC–AUC. A comparative study of diagnostic performance, robustness, and computational efficiency is then conducted to identify the most suitable model for reliable and practical clinical deployment.

COMPARE ALGORITHMS

 Algorithm	 Works on Raw Images?	 Accuracy	 Speed	 Suitable For
 Decision Tree	No	Moderate	Fast	Tabular data
 KNN	No	Moderate	Slow	Small datasets
 SVM	No	Good	Medium	Small feature sets
 Pre-trained CNN (ResNet50, DenseNet121, VGG16, MobileNetV2)	Yes	Very Good	Medium	Small medical datasets
  Custom CNN (Our Model)	Yes	Good	Medium	Image classification



Selected Algorithm

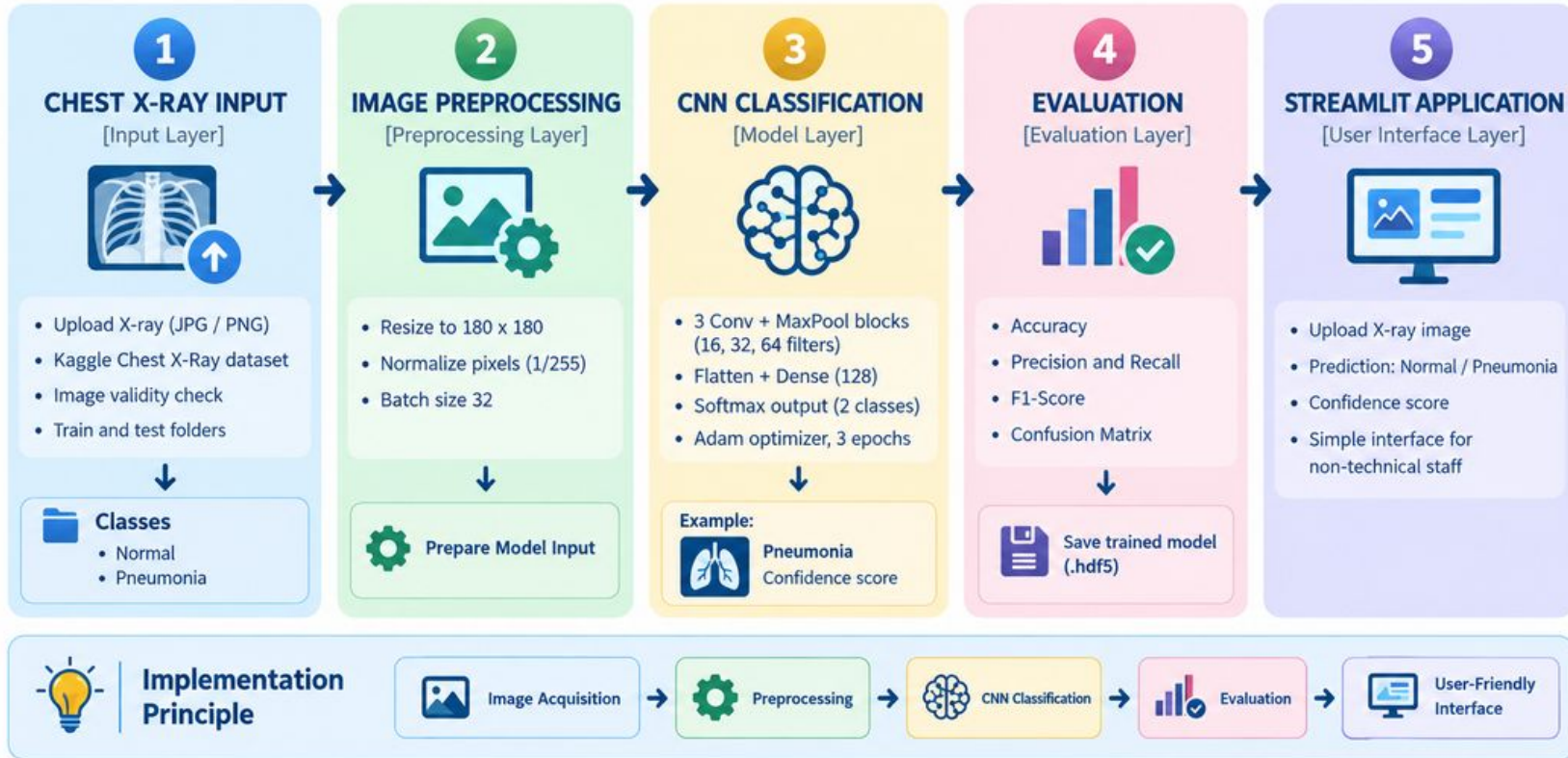


Custom CNN

Why CNN?

- ✓ Learns features directly from X-ray pixels
- ✓ Keeps spatial patterns like opacity and texture
- ✓ Standard baseline for image classification
- ✓ Simple to train and deploy

IMPLEMENTATION IDEA



MODULE DESCRIPTION

1. Image Input Module

Allows users (doctors or staff) to upload chest X-ray images into the system through a simple interface. Supports common image formats (JPEG, PNG) and performs a basic check to ensure the uploaded file is a valid X-ray image before passing it to the next stage.

2. Preprocessing Module

Prepares the raw X-ray image for the model by resizing it to a fixed input size, normalizing pixel values, and applying basic enhancements (e.g., contrast adjustment) if needed. This ensures consistency across images from different sources and improves model accuracy.

MODULE DESCRIPTION

3. Feature Extraction Module

Uses a pre-trained deep learning model (e.g., ResNet) to analyze the preprocessed X-ray and extract important visual patterns, such as lung opacity and texture, that are relevant to identifying pneumonia. This module leverages knowledge the model already learned from large image datasets, reducing the need for extensive training data.

4. Classification Module

Takes the extracted features and passes them through a final classification layer to predict whether the X-ray shows signs of pneumonia or appears normal. Outputs a prediction along with a confidence score indicating how certain the model is about its result.



MODULE DESCRIPTION



5. Result Display Module

Presents the classification result to the user in a clear, easy-to-understand format, showing the diagnosis (Pneumonia/Normal) along with the confidence score. Designed to be simple enough for non-technical hospital staff to interpret quickly.

6. Data Storage Module

Stores the uploaded X-ray images, along with their corresponding results and timestamps, in a database for record-keeping. This allows for future reference, tracking patient history, or further analysis to improve the model over time.



CONCLUSION



- Reasonable classification accuracy on a test set of chest X-ray images.
- Faster turnaround time for preliminary screening compared to fully manual review.
- A working prototype that classifies chest X-ray images as "Pneumonia" or "Normal."
- A simple, functional interface that requires no technical expertise to operate.
- A foundation that can be improved further with more data, tuning, or additional features.

THANK YOU